Exploratory Data Analysis for SPROJ

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Load Packages

In order to do this Exploratory Data Analysis, I will use the following packages:

```
library(ggplot2) # for plotting data
library(readxl) # for reading excel files if necessary
library(dplyr) # for data wrangling
# library(arrow)
library(janitor) # for cleaning messy names of datasets
library(readr) # for reading files
library(gt) # for making 'publication ready' tables
library(usethis) #enables git pushing to put on github
library(gitcreds)
library(purrr) # this package is to loop and automate this entire process, instead of going t
```

This senior project document will serve as the backbone of methodology and analysis part of the Senior Project"

Motivation

According to the UN Trade Hub and Development:

(https://unctadstat.unctad.org/datacentre/reportInfo/US.RCA)

"Revealed comparative advantage (RCA) is based on Ricardian trade theory, which posits that patterns of trade among countries are governed by their relative differences in productivity. Although such productivity differences are difficult to observe, an RCA metric can be readily calculated using trade data to"reveal" such differences. While the metric can be used to provide a general indication and first approximation of a country's competitive export strengths, it should be noted that applied national measures which affect competitiveness such as tariffs, non-tariff measures, subsidies and others are not taken into account in the RCA metric."

"When a country has a revealed comparative advantage for a given product (RCA >1), it is inferred to be a competitive producer and exporter of that product relative to a country producing and exporting that good at or below the world average. A country with a revealed comparative advantage in product i is considered to have an export strength in that product. The higher the value of a country's RCA for product i, the higher its export strength in product i."

Part of this senior project aims to explain shifts in Revealed Comparative Advantage through a political economic lens, specifically in Mexico's Agriculture industry As the UN mentions, this RCA is merely an inference variable, and is used as a 'first approximation' in comparative advantage. Many of the shifts in RCA over time are not captured. I will use classical economic theory, neoclassical theory, economic literature, and papers to describe shifts in the RCA through a political economic lens. My sector analysis in these documents will be done in order of sector code and number, starting from 01; Animals Live, throughout the rest of the agricultural sector.

The agricultural sector is one that is extremely broad- ranging from plants and livestock, to coffee beans and even honey. In order to further analyze the RCA index, we must discern what codes constitute the agriculture industry. Each sector has its own general product code at the 2 digit, 4 digit, and 6 digit level, with each digits being more specific in each product. Using the International Trade Administration (ITA) Harmonized System (HS) codes, we will use these definitions to discern what products are part of the agriculture industry. There is opinions in the literature that showws variations in the product codes that is part of the 'agricultural sector.' I will use the definitions from this paper (https://www.researchgate.net/publication/332551060_Changes_in_Foreign_Trade_in_Agricultural_Produc to classify the codes that belongs in the agricultural sector. Furthermore, I will analyze each sector at a general level, starting at the 2 digit, and then up to 4 digits. Due to constraints and limitations of the publicly available data, the maximum scope of analysis is up to the 4 digit HS code level. The data will also be split up into multiple time periods. Period A will be from 1992-2000, period B will be from 2000-2008, and period C will be from 2008-2016, period D will be from 2016-2024.

Functions

The section below describes the functions which are necessary in computing this analysis, I have made comments in the code chunks in order to further expand on each line of code and what it does.

Necessary Variables Function

```
## This is a function that takes in only the necessary variables for an RCA analysis, includ

necessary_variables <- function(df) { # inputs a dataframe
  df |>
    dplyr::mutate(cmdCode = as.character(cmdCode)) |>
    arrange(period) |> #arranges the df by year
    select(period, reporterISO, cmdCode, flowCode, qty, netWgt, fobvalue, primaryValue) #sele
}
```

RCA Function

```
## RCA calculation function

#inputs country exports of product i , country's total exports, world exports of product i, a
calculate_rca <- function(country_product, country_total, world_product, world_total) {

  #join by period to align years
  rca_df <- country_product |>
    inner_join(country_total, by = "period", suffix = c("_prod", "_total")) |>
    inner_join(world_product, by = "period") |>
    inner_join(world_total, by = "period", suffix = c("_world_prod", "_world_total")) |>
    mutate(
      RCA = (fobvalue_prod / fobvalue_total) /
            (fobvalue_world_prod / fobvalue_world_total)
    ) |>
    select(period, RCA)

  return(rca_df)
}
```

Generalized RCA Function

```
calculate_rca_generalized <- function(country_sector_df,
                                      world_sector_df,
                                      country_all_df,
                                      world_all_df) {
```

```r
  #first, I aggregate by year and product, with variables period, and cmdCode, removing any n
  country_prod <- country_sector_df |>
    group_by(period, cmdCode) |>
    summarise(country_product_value = sum(fobvalue, na.rm = TRUE), .groups = "drop")

  world_prod <- world_sector_df |>
    group_by(period, cmdCode) |>
    summarise(world_product_value = sum(fobvalue, na.rm = TRUE), .groups = "drop")

  #next, I aggregate total exports using fobvalue  (denominator)
  country_all <- country_all_df |>
    group_by(period) |>
    summarise(total_country_value = sum(fobvalue, na.rm = TRUE), .groups = "drop")

  world_all <- world_all_df |>
    group_by(period) |>
    summarise(total_world_value = sum(fobvalue, na.rm = TRUE), .groups = "drop")

  # 3. next, I merge everytjing by year and commodity code
  rca_df <- country_prod |>
    left_join(world_prod, by = c("period", "cmdCode")) |>
    left_join(country_all, by = "period") |>
    left_join(world_all, by = "period") |>
    mutate(
      RCA = (country_product_value / total_country_value) /
            (world_product_value / total_world_value),
      has_CA = RCA >= 1
    ) |>
    arrange(period, cmdCode)

  return(rca_df)
}
```

Necessary Dataframes

```r
#mexico total exports time 1992 -> 2000

mx_xports_all <- read_csv("mx_total_xports_world/mx_xports_all.csv")
```

```r
## world exports of 07 time 1992 -> 2000

world_xports_all <- read_csv("world_total_xports/world_xports_all.csv")
```

```r
mx_xports_all <- necessary_variables(mx_xports_all) # dataframe for all of mexico's exports
world_xports_all <- necessary_variables(world_xports_all) # dataframe for world exports of al
```

We need the dataframes above in order to complete the rest of the RCA Analysis. They are imperative in order so that the RCA variable can be calculated at a snapshot in time.

Loop Initiation

```r
## create sector codes

# sector_codes <- seq(1:99) #creates a list of integers from 1 to 99

# sector_codes <- sprintf("%02d", 1:99) # since it is stored as 01..99 this adds a 01 for all

#in this instance, I will manually add the codes, as I will not be analyzing all sectors. The

sector_codes <- c("01", "02", "04", "05","07","08","09","10","11")
```

process 1 Sector

```r
process_sector <- function(sector) { ## sector in this instance is HS classification code

  mx_file <- paste0("mx_xports_prod_i/mx_xports_sector_", sector, ".csv") ## paste0 is used
  world_file <- paste0("world_xports_prod_i/world_xports_sector_", sector, ".csv") ## simila

  # skip sector if files do not exist, if file does not exist in mx_file or file does not ex
  if (!file.exists(mx_file) | !file.exists(world_file)) {
    message("Skipping sector ", sector)
    return(NULL)
  }

  mx_sector <- read_csv(mx_file, show_col_types = FALSE) |> necessary_variables()
```

```
  world_sector <- read_csv(world_file, show_col_types = FALSE) |> necessary_variables()

  rca_sector <- calculate_rca_generalized(
    mx_sector,
    world_sector,
    mx_xports_all,
    world_xports_all
  )

  rca_sector$sector <- sector

  return(rca_sector)
}
```

01 Animals; Live

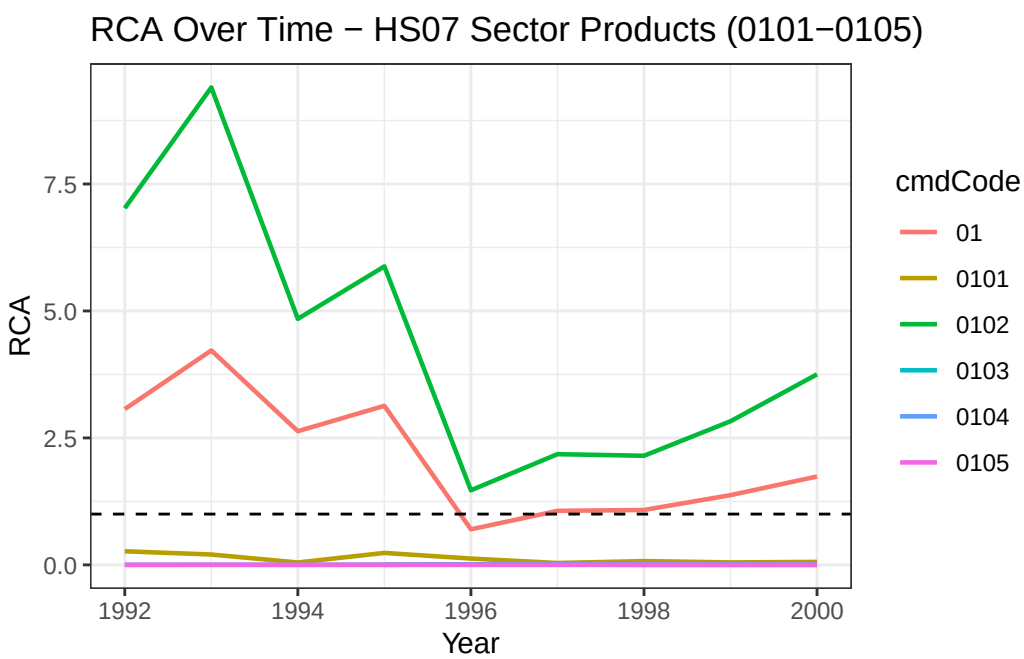
This section will calculate RCA over time before NAFTA of the animal section, containing HS product code of 0101 to 0106.

```
## load the datasets
mx_xports_sector_01 <- read_csv("mx_xports_prod_i/mx_xports_sector_01.csv")
world_xports_sector_01 <- read_csv("world_xports_prod_i/world_xports_sector_01.csv")


## necessary variables

mx_xports_sector_01 <- necessary_variables(mx_xports_sector_01)
world_xports_sector_01 <- necessary_variables(world_xports_sector_01)


# calculate RCA


rca_sector_01 <- calculate_rca_generalized(mx_xports_sector_01, world_xports_sector_01, mx_xp
```

ggplot sector 01 from 1992 - 2000.

```
ggplot(rca_sector_01, aes(x = period, y = RCA, color = cmdCode, group = cmdCode)) +
  geom_line(size = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed") +
```

```
  labs(title = "RCA Over Time - HS07 Sector Products (0101-0105)",
       x = "Year", y = "RCA") +
  theme_bw()
```

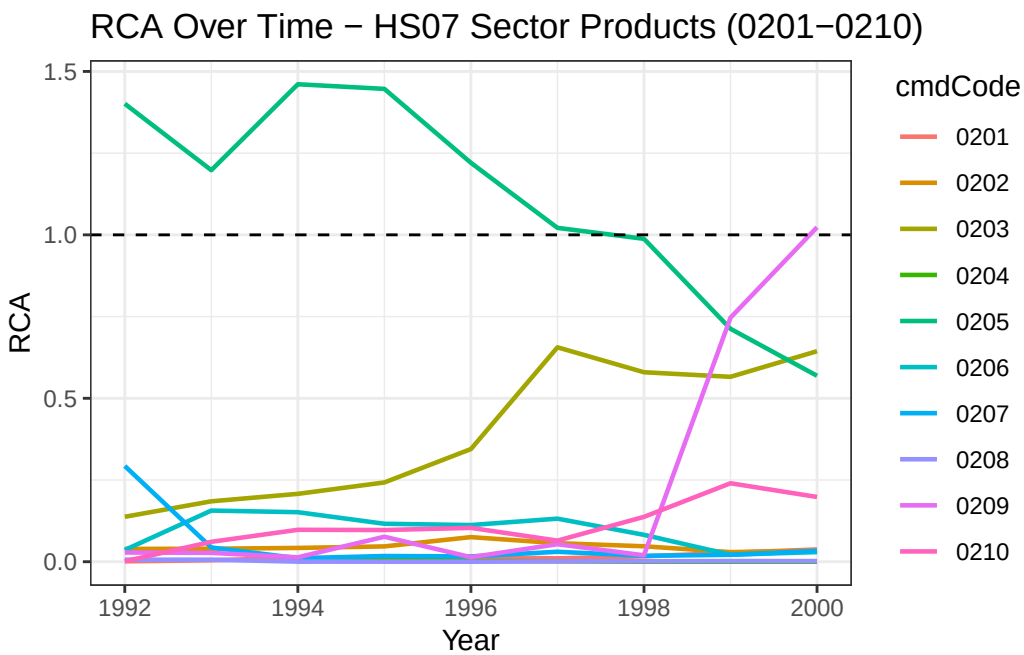


Interesting that Mexico seems to have a general RCA of Live Animals, but there was a sharop decline starting in 1993. NAFTA was enacted January 1, 1994- was it a driving factor in RCA shift? What about data from future years?

02 Meat and Edible Meat offal

```
## load the dataframes
mx_xports_sector_02 <- read_csv("mx_xports_prod_i/mx_xports_sector_02.csv")
world_xports_sector_02 <- read_csv("world_xports_prod_i/world_xports_sector_02.csv")

# overwrite with necessary variables

mx_xports_sector_02 <- necessary_variables(mx_xports_sector_02)
world_xports_sector_02 <- necessary_variables(world_xports_sector_02)

## calculate rca
rca_sector_02 <- calculate_rca_generalized(mx_xports_sector_02, world_xports_sector_02, mx_xp
```

```r
## graph RCA
ggplot(rca_sector_02, aes(x = period, y = RCA, color = cmdCode, group = cmdCode)) +
  geom_line(size = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed") +
  labs(title = "RCA Over Time - HS07 Sector Products (0201-0210)",
       x = "Year", y = "RCA") +
  theme_bw()
```



Sector of 02 seems to not have much of a comparative advantage in Mexico.

04 Dairy Produce; Birds' eggs; Natural Honey; Edible products of animal origin

04, 0401 -> 0410 EXCLUDING HONEY 0409

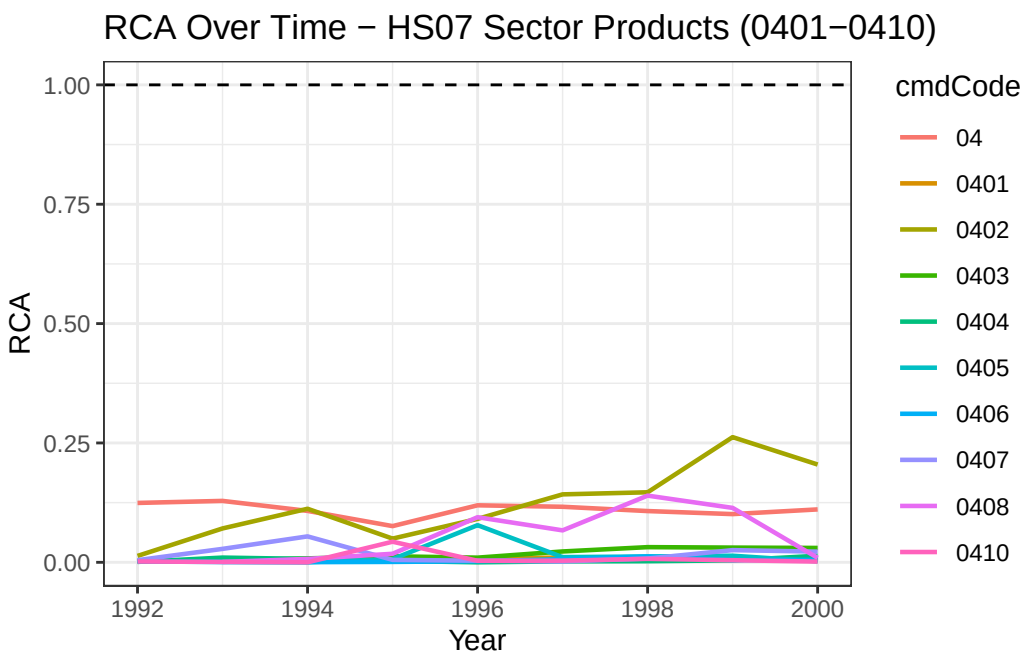
First, we load the dataframes necessary, and overwrite them with necessary variables

```r
mx_xports_sector_04 <- read_csv("mx_xports_prod_i/mx_xports_sector_04.csv")

world_xports_sector_04 <- read_csv("world_xports_prod_i/world_xports_sector_04.csv")

mx_xports_sector_04 <- necessary_variables(mx_xports_sector_04)
world_xports_sector_04 <- necessary_variables(world_xports_sector_04)
```

Calculate RCA and then graph

```
rca_sector_04 <- calculate_rca_generalized(mx_xports_sector_04, world_xports_sector_04, mx_xp
```

```
ggplot(rca_sector_04, aes(x = period, y = RCA, color = cmdCode, group = cmdCode)) +
  geom_line(size = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed") +
  labs(title = "RCA Over Time - HS07 Sector Products (0401-0410)",
       x = "Year", y = "RCA") +
  theme_bw()
```



05 0ther Animal Products

ANIMAL ORIGINATED PRODUCTS; NOT ELSEWHERE SPECIFIED OR INCLUDED

In this sector, I will avoid

- 0501 Human hair; unworked, whether or not washed or scoured; waste of human hair
- 0509 Sponges

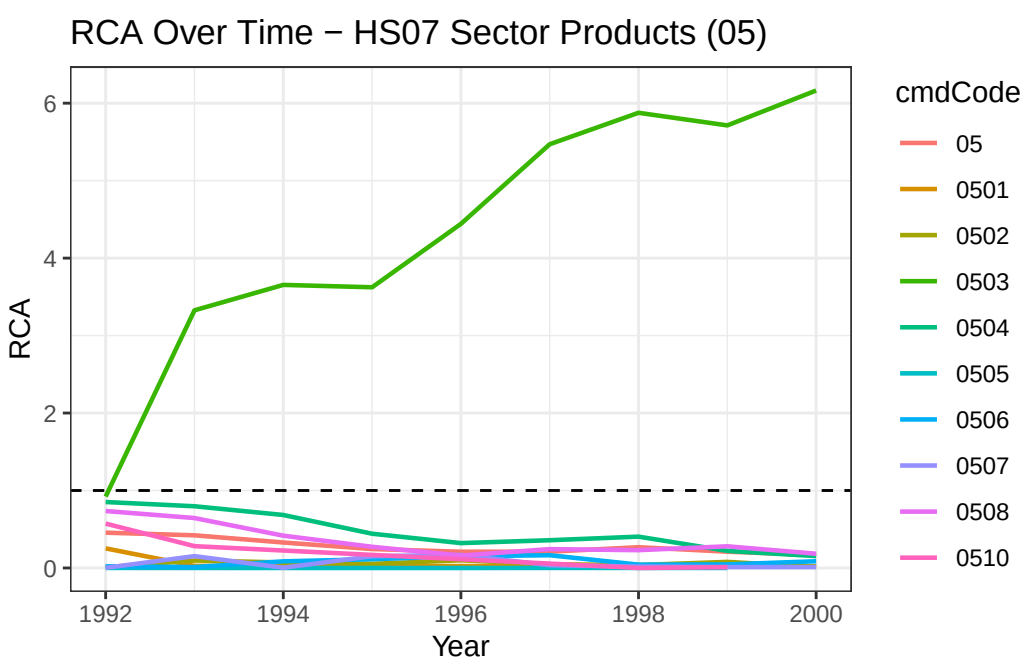
- 0511 Animal products not elsewhere specified or included; dead animals of chapter 1 or 3, unfit for human consumption

```
mx_xports_sector_05 <- read_csv("mx_xports_prod_i/mx_xports_sector_05.csv")

world_xports_sector_05 <- read_csv("world_xports_prod_i/world_xports_sector_05.csv")

mx_xports_sector_05 <- necessary_variables(mx_xports_sector_05)
world_xports_sector_05 <- necessary_variables(world_xports_sector_05)
```

Calculate RCA and then graph

```
rca_sector_05 <- calculate_rca_generalized(mx_xports_sector_05, world_xports_sector_05, mx_xp
```

```
ggplot(rca_sector_05, aes(x = period, y = RCA, color = cmdCode, group = cmdCode)) +
  geom_line(size = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed") +
  labs(title = "RCA Over Time - HS07 Sector Products (05)",
       x = "Year", y = "RCA") +
  theme_bw()
```



Interestingly, 0503 "Horsehair, waste" has a comparative advantage in that product, but the sector does not have a comparative advantage.

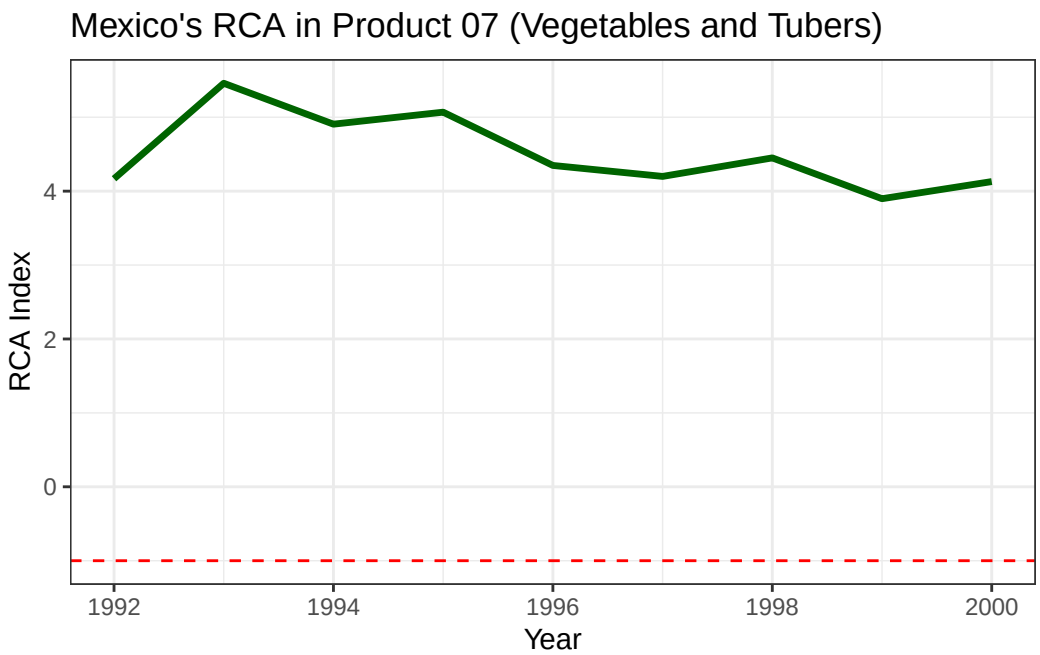
07- Vegetables and Certain Roots and Tubes; Edible

```
# read the datasets in each folder

#mexico exports of 07 VEGETABLES AND CERTAIN ROOTS AND TUBERS; EDIBLE time 1992 -> 2000
mx_xports_07 <- read_csv("mx_xports_prod_i/mx_xports_07.csv")

world_xports_07 <- read_csv("world_xports_prod_i/world_xports_07.csv")
```

```
mx_xports_07 <- necessary_variables(mx_xports_07)
mx_xports_all <- necessary_variables(mx_xports_all)
world_xports_07 <- necessary_variables(world_xports_07)
world_xports_all <- necessary_variables(world_xports_all)
```

```
## apply to datafranes

mx_rca_07_1992_2000 <- calculate_rca(
  mx_xports_07,
  mx_xports_all,
  world_xports_07,
  world_xports_all)
```

```
ggplot(mx_rca_07_1992_2000, aes(x = period, y = RCA)) +
  geom_line(color = "darkgreen", size = 1.2) +
  geom_hline(yintercept = -1, linetype = "dashed", color = "red") +
  labs(
    title = "Mexico's RCA in Product 07 (Vegetables and Tubers)",
    x = "Year",
    y = "RCA Index"
  ) +
  theme_bw()
```



Sector Analysis : 07

Now, I are tasked to clean the dataset and calculate an RCA given the dataset with the entire sector. Using the code above, find a way to generalize the RCA calculation to continue the calculation, with an ultimate goal of showing an RCA graph for each product in the sector.

```
# read the dataframe of the 07 sector contaning country MX exports of product 0701, 0702, 070

#mexico exports of products 0701 to 0710 and time 1992 -> 2000

mx_xports_sector_07 <- read_csv("mx_xports_prod_i/mx_xports_sector_07.csv")
```

Now, I must use the necessary variables for the RCA. necessary variables are the following: period, reporterISO, cmdCode, flowCode, qty, netWgt, fobvalue, primaryValue. Recall that the RCA variable is calculated with the following:

RCA = [(Country Exports of Product i / Country A Exports of ALL products) / (World Exports of product i / World Exports of all products)]

$$\text{RCA}_i = \frac{\frac{X_{A,i}}{X_{A,\text{all}}}}{\frac{X_{World,i}}{X_{World,\text{all}}}}$$

The RCA variable calculates a share of country exports a product over the world exports of said product. If the RCA variable is greater than, or equal to 1, then the country has a revealed comparative advantage in producing said good. According to neoclassical theory, gains from trade are still probable in comparative advantage. Theory says to export goods in which the country has a comparative advantage in producing.

```
mx_xports_sector_07 <- necessary_variables(mx_xports_sector_07)
```

Now, I have the necessary variables, Country MX Exports of product 0701 -> 0710. I have the data of Mexico's exports of all products, titled `mx_xports_all`. Now, I need World's exports of products 0701 -> 0710. I can do this by using UNCOMTRADE data as Ill. Repeating the same steps, I will grab the necessary variables.

```
#load the dataframe of world exports of prod 0701 -> 0710
world_xports_sector_07 <- read_csv("world_xports_prod_i/world_xports_sector_07.csv")

#grab necessary variables.
world_xports_sector_07 <- necessary_variables(world_xports_sector_07)
```

I now have World exports of product 0701 -> 0710 with the same time period of 1992 to 2000. Recall I have datasets of Mexico's exports of all products titled `mx_xports_all` with variables period, reporterISO, cmdCode, flowCode, qty, netWgt, fobvalue, primaryValue. I also have `world_xports_all` which contains world exports of all products with variables period, reporterISO, cmdCode, flowCode, qty, netWgt, fobvalue, primaryValue. HoIver, notice that `world_xports_sector_07` and `mx_xports_sector_07` have the cmdCode, hoIver, they are not aggregated, and their cmdCode varies from 0701 -> 0710. Below is an attempted geenralized calclation of RCA, with a clear difference of grouping by `group_by(period, cmdCode)` due to the nature of the dataframe with varying cmdCode

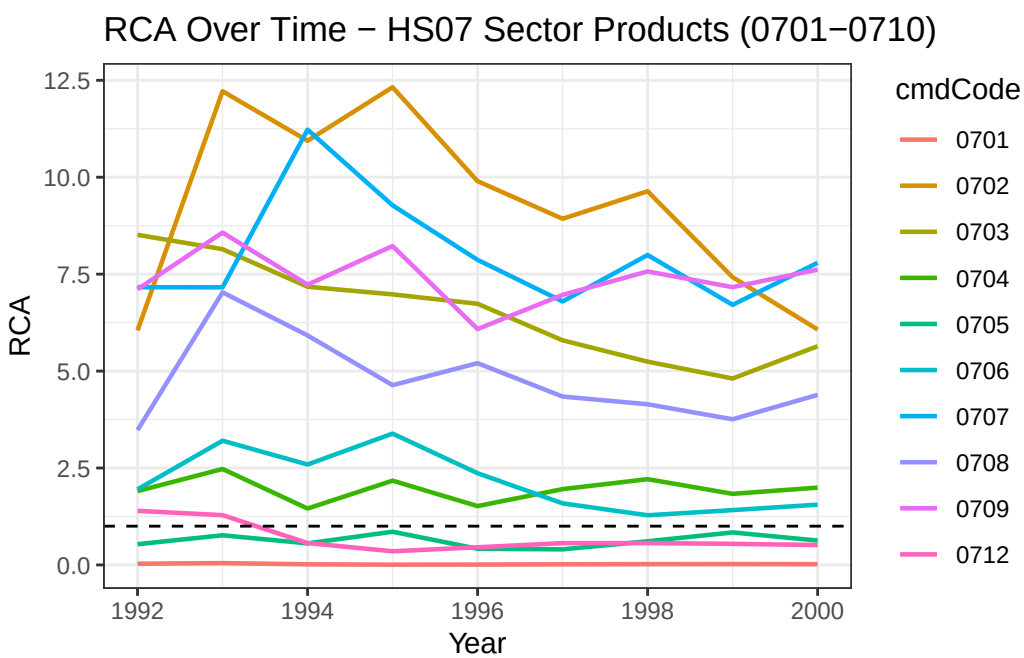
Let's test this in action to see if it works!

RCA of sector 0701 to 0710.

```
# inputs need: country_sector_df, world_sector_df, country_all_df, world_all_df

# have: mx_xports_sector_07, world_xports_sector_07, mx_xports_all, world_xports_all

rca_sector_07 <- calculate_rca_generalized(mx_xports_sector_07, world_xports_sector_07,mx_xpo
```

Visualize over time:

```r
## ggplot to bisualize over time
ggplot(rca_sector_07, aes(x = period, y = RCA, color = cmdCode, group = cmdCode)) +
  geom_line(size = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed") +
  labs(title = "RCA Over Time - HS07 Sector Products (0701-0710)",
       x = "Year", y = "RCA") +
  theme_bw()
```



I can now develop code to show which products have an RCA > 1 and RCA < 1.

I can do this by running a simple filter:

```r
## show which years had an RCA > 1
table_RCA_gt1 <- rca_sector_07 |>
  filter(RCA > 1) |>
  arrange(period, cmdCode)
table_RCA_gt1
```

```
# A tibble: 65 x 8
   period cmdCode country_product_value world_product_value total_country_value
    <dbl> <chr>                   <dbl>               <dbl>               <dbl>
 1   1992 0702                166984992          1467571657         184779539480
 2   1992 0703                123081000           768524226         184779539480
 3   1992 0704                 14027000           391613956         184779539480
```

```
 4   1992 0706                 10529000           287038881        184779539480
 5   1992 0707                 91608000           679846342        184779539480
 6   1992 0708                 16944000           258995555        184779539480
 7   1992 0709                302278016          2260100293        184779539480
 8   1992 0712                 17868000           680654076        184779539480
 9   1993 0702                394972000          1762452605        207545660288
10   1993 0703                146144000           978346042        207545660288
# i 55 more rows
# i 3 more variables: total_world_value <dbl>, RCA <dbl>, has_CA <lgl>
```

or by showing which years had an RCA > 1

```
#summarize the comparative advantage by product by grouping by product code
summary_CA_by_product <- rca_sector_07 |>
  group_by(cmdCode) |>
  summarise(
    years_with_CA = paste(period[RCA > 1], collapse = ", "),
    n_years_CA = sum(RCA > 1),
    total_years = n(),
    share_years_CA = n_years_CA / total_years
  )
```

and a publication ready table:

```
summary_CA_by_product |>
  gt() |>
  tab_header(
    title = "Mexico's RCA, Sector 07 (RCA > 1)",
  )
```

I can fix this dataframe by changing the name of the previous one

Sector definitions

07 - VEGETABLES AND CERTAIN ROOTS AND TUBERS; EDIBLE

- 0701 - Potatoes; fresh or chilled
- 0702 - Tomatoes; fresh or chilled
- 0703 - Onions, shallots, garlic, leeks and other alliaceous vegetables; fresh or chilled

[

Mexico's RCA, Sector 07 (RCA > 1)] Mexico's RCA, Sector 07 (RCA > 1)

cmdCode	years_with_CA	n_years_CA	total_years	share_y
0701		0	9	0
0702	1992, 1993, 1994, 1995, 1996, 1997, 1998, 1999, 2000	9	9	
0703	1992, 1993, 1994, 1995, 1996, 1997, 1998, 1999, 2000	9	9	
0704	1992, 1993, 1994, 1995, 1996, 1997, 1998, 1999, 2000	9	9	
0705		0	9	0
0706	1992, 1993, 1994, 1995, 1996, 1997, 1998, 1999, 2000	9	9	
0707	1992, 1993, 1994, 1995, 1996, 1997, 1998, 1999, 2000	9	9	
0708	1992, 1993, 1994, 1995, 1996, 1997, 1998, 1999, 2000	9	9	
0709	1992, 1993, 1994, 1995, 1996, 1997, 1998, 1999, 2000	9	9	
0712	1992, 1993	2	9	0

- 0704 - Cabbages, caulifloIrs, kohlrabi, kale and similar edible brassicas; fresh or chilled
- 0705 - Lettuce (lactuca sativa) and chicory (cichorium spp.) fresh or chilled
- 0706 - Carrots, turnips, salad beetroot, salsify, celeriac, radishes and similar edible roots; fresh or chilled
- 0707 - Cucumbers and gherkins; fresh or chilled
- 0708 - Leguminous vegetables; shelled or unshelled, fresh or chilled
- 0709 - Vegetables; n.e.c. in chapter 07, fresh or chilled
- 0710 - Vegetables (uncooked or cooked by steaming or boiling in water); frozen
- 0711 - Vegetables provisionally preserved; (e.g. by sulphur dioxide gas, in brine, in sulphur water or in other preservative solutions), but unsuitable in that state for immediate consumption
- 0712 - Vegetables, dried; whole, cut, sliced, broken or in powder, but not further prepared
- 0713 - Vegetables, leguminous; shelled, whether or not skinned or split, dried
-

Today Kash helped me tremendously setting up git hub for the first time ever!!

08- FRUIT AND NUTS, EDIBLE; PEEL OF CITRUS FRUIT OR MELONS

We omit the following:

- 0812 Fruit and nuts provisionally preserved; e.g. by sulphur dioxide gas, brine, in sulphur water or in other preservative solutions, but unsuitable in that state for immediate consumption
- 0813 Fruit, dried, other than that of heading no. 0801 to 0806; mixtures of nuts or dried fruits of this chapter
- 0814 Peel of citrus fruit or melons (including watermelons); fresh, frozen dried or provisionally preserved in brine, in sulphur water or in other preservative solutions

```
mx_xports_sector_08 <- read_csv("mx_xports_prod_i/mx_xports_sector_08.csv")

world_xports_sector_08 <- read_csv("world_xports_prod_i/world_xports_sector_08.csv")

mx_xports_sector_05 <- necessary_variables(mx_xports_sector_08)
world_xports_sector_05 <- necessary_variables(world_xports_sector_08)
```

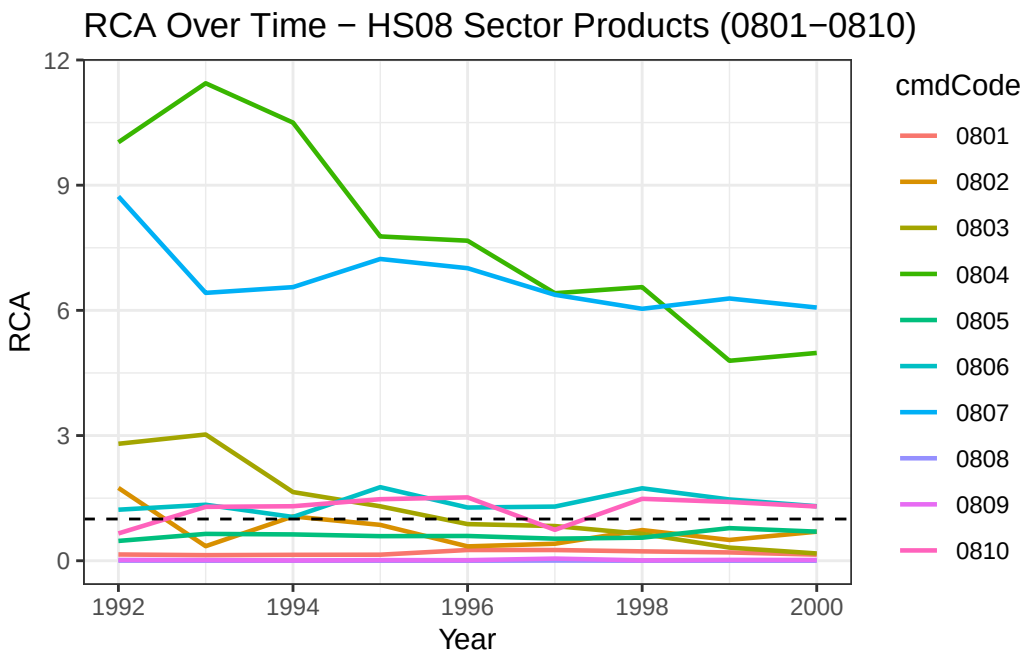
RCA of sector 0801 to 0810.

```
# inputs need: country_sector_df, world_sector_df, country_all_df, world_all_df

# have: mx_xports_sector_07, world_xports_sector_07, mx_xports_all, world_xports_all

rca_sector_08 <- calculate_rca_generalized(mx_xports_sector_08, world_xports_sector_08,mx_xpo
```

Plot sector 08

Visualize over time:

```
## ggplot to bisualize over time
ggplot(rca_sector_08, aes(x = period, y = RCA, color = cmdCode, group = cmdCode)) +
  geom_line(size = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed") +
  labs(title = "RCA Over Time - HS08 Sector Products (0801-0810)",
       x = "Year", y = "RCA") +
  theme_bw()
```



09- COFFEE, TEA, MATE AND SPICES

No 4 digit HS classification code is omitted.

```
mx_xports_sector_09 <- read_csv("mx_xports_prod_i/mx_xports_sector_09.csv")

world_xports_sector_09 <- read_csv("world_xports_prod_i/world_xports_sector_09.csv")

mx_xports_sector_09 <- necessary_variables(mx_xports_sector_09)
world_xports_sector_09 <- necessary_variables(world_xports_sector_09)
```

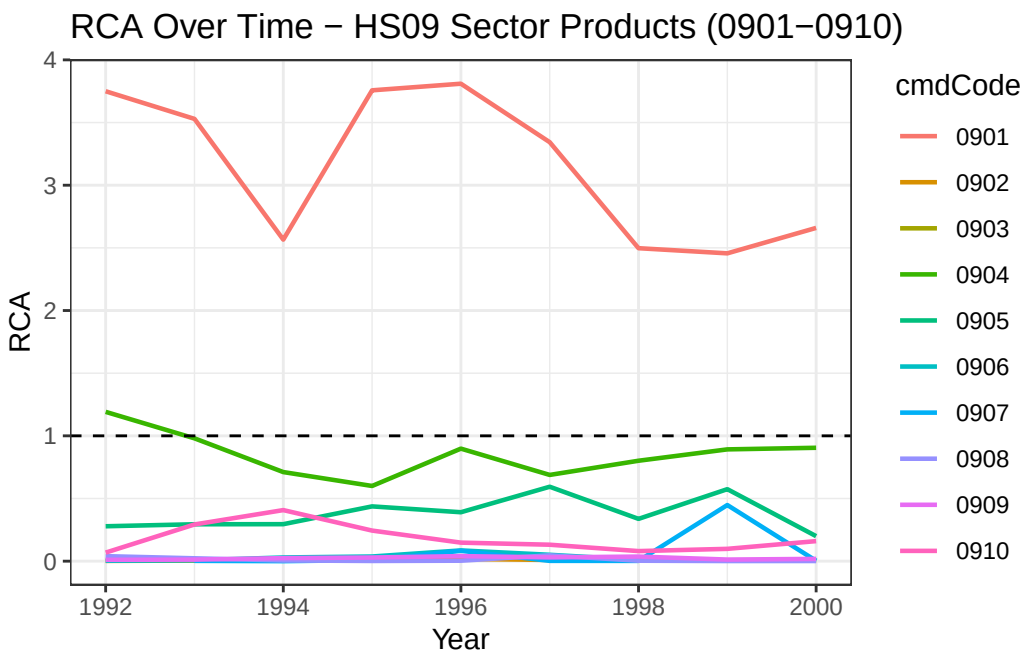
RCA of sector 0901 to 0910.

```
# inputs need: country_sector_df, world_sector_df, country_all_df, world_all_df

# have: mx_xports_sector_07, world_xports_sector_07, mx_xports_all, world_xports_all

rca_sector_09 <- calculate_rca_generalized(mx_xports_sector_09, world_xports_sector_09,mx_xpo
```

Plot sector 09

Visualize over time:

```r
## ggplot to bisualize over time
ggplot(rca_sector_09, aes(x = period, y = RCA, color = cmdCode, group = cmdCode)) +
  geom_line(size = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed") +
  labs(title = "RCA Over Time - HS09 Sector Products (0901-0910)",
       x = "Year", y = "RCA") +
  theme_bw()
```



10 CEREALS (needs fix, ask Prof. Valerie)

No 4 digit HS classification code is omitted.

```r
mx_xports_sector_10 <- read_csv("mx_xports_prod_i/mx_xports_sector_10.csv")

world_xports_sector_10 <- read_csv("world_xports_prod_i/world_xports_sector_10.csv")

mx_xports_sector_10 <- necessary_variables(mx_xports_sector_10)
world_xports_sector_10 <- necessary_variables(world_xports_sector_10)
```

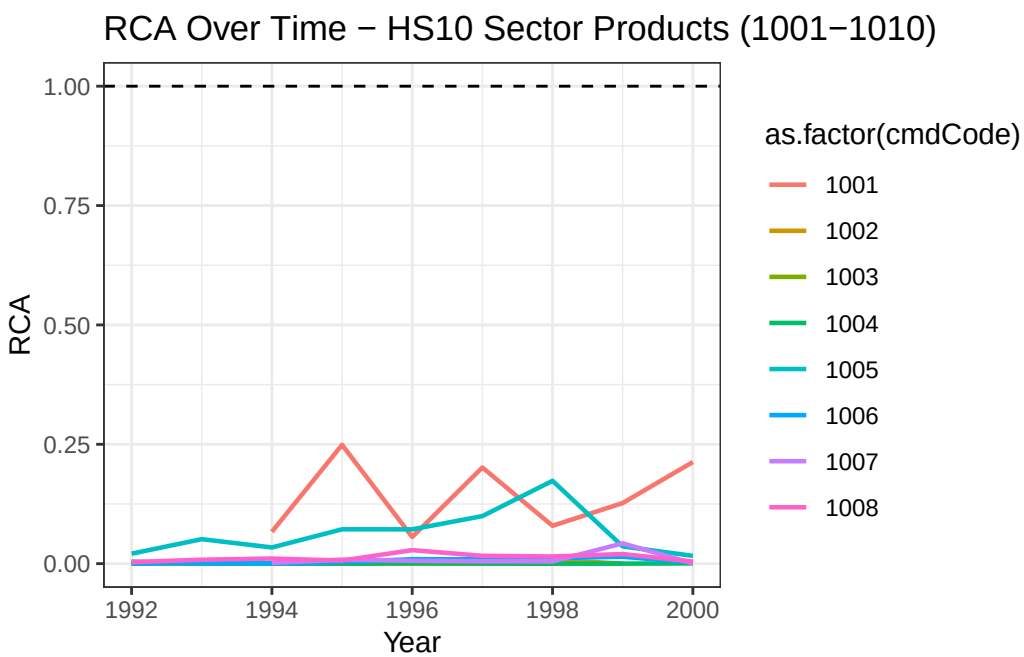
RCA of sector 1001 to 1008

```
# inputs need: country_sector_df, world_sector_df, country_all_df, world_all_df

# have: mx_xports_sector_10, world_xports_sector_10, mx_xports_all, world_xports_all

rca_sector_10 <- calculate_rca_generalized(mx_xports_sector_10, world_xports_sector_10,mx_xpo
```

Plot sector 10

Visualize over time:

```
## ggplot to bisualize over time
ggplot(rca_sector_10, aes(x = period, y = RCA, color = as.factor(cmdCode), group = cmdCode))
  geom_line(size = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed") +
  labs(title = "RCA Over Time - HS10 Sector Products (1001-1010)",
       x = "Year", y = "RCA") +
  theme_bw()
```



11 Milling products; malt; starch, etc.; gluten

No 4 digit HS classification code is omitted.

```r
mx_xports_sector_11 <- read_csv("mx_xports_prod_i/mx_xports_sector_11.csv")

world_xports_sector_11 <- read_csv("world_xports_prod_i/world_xports_sector_11.csv")

mx_xports_sector_11 <- necessary_variables(mx_xports_sector_11)
world_xports_sector_11 <- necessary_variables(world_xports_sector_11)
```

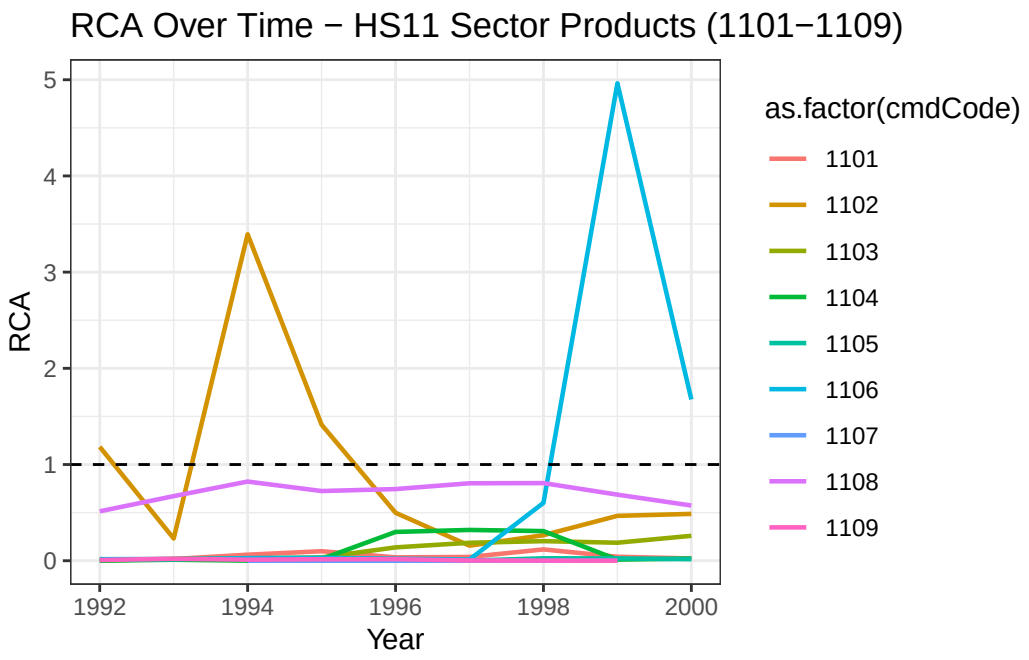
RCA of sector 1101 to 1109

```r
# inputs need: country_sector_df, world_sector_df, country_all_df, world_all_df

# have: mx_xports_sector_10, world_xports_sector_10, mx_xports_all, world_xports_all

rca_sector_11 <- calculate_rca_generalized(mx_xports_sector_11, world_xports_sector_11,mx_xpo
```

Plot sector 11

Visualize over time:

```r
## ggplot to bisualize over time
ggplot(rca_sector_11, aes(x = period, y = RCA, color = as.factor(cmdCode), group = cmdCode))
  geom_line(size = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed") +
  labs(title = "RCA Over Time - HS11 Sector Products (1101-1109)",
       x = "Year", y = "RCA") +
  theme_bw()
```



12 Oilseeds; kernels; industrial medicinal plants; feed

We omit the following HS Codes:

- 1205 Rape or colza seeds; whether or not broken
- 1211 Plants and parts of plants (including seeds and fruits), of a kind used primarily in perfumery, in pharmacy or for insecticidal, fungicidal or similar purposes, fresh, chilled, frozen or dried, whether or not cut, crushed or powdered
- 1212 Locust beans, seaweeds and other algae, sugar beet, sugar cane, fresh, chilled, frozen or dried, whether or not ground; fruit stones, kernels and other vegetable products (including unroasted chicory roots) used primarily for human consumption, n.e.c.
- 1214 Swedes, mangolds, fodder roots, hay, lucerne (alfalfa), clover, sainfoin, forage kale, lupines, vetches and similar forage products, whether or not in the form of pellets

No 4 digit HS classification code is omitted.

```
mx_xports_sector_12 <- read_csv("mx_xports_prod_i/mx_xports_sector_12.csv")

world_xports_sector_12 <- read_csv("world_xports_prod_i/world_xports_sector_12.csv")

mx_xports_sector_12 <- necessary_variables(mx_xports_sector_12)
```

```r
world_xports_sector_12 <- necessary_variables(world_xports_sector_12)

mx_xports_sector_12 <- mx_xports_sector_12 %>%
  mutate(fobvalue = as.numeric(as.character(fobvalue)))

world_xports_sector_12 <- world_xports_sector_12 %>%
  mutate(fobvalue = as.numeric(as.character(fobvalue)))
```

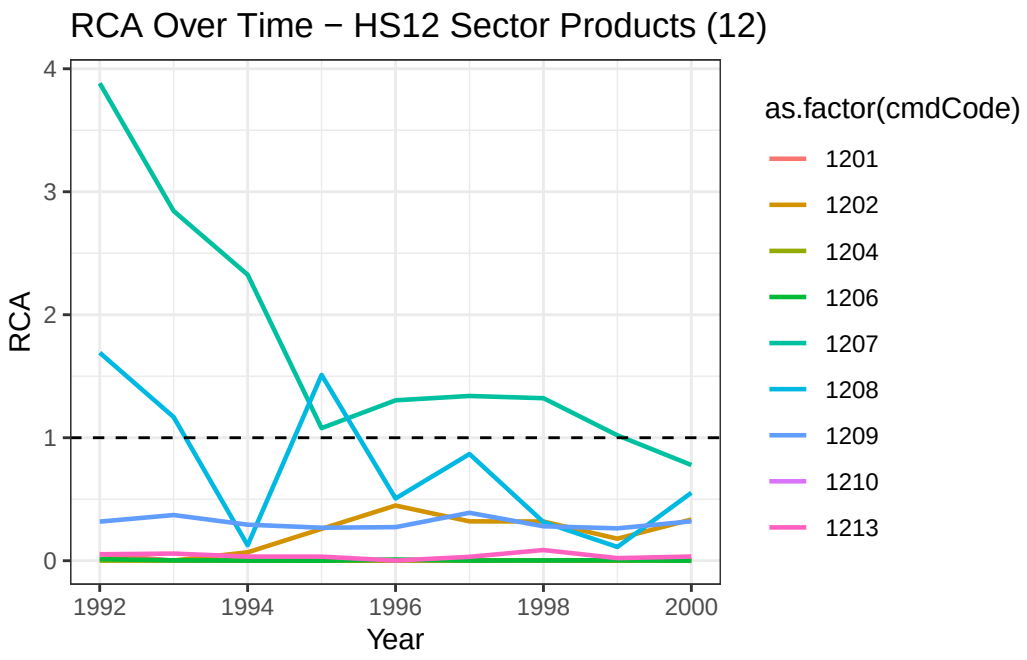
RCA of sector 12

```r
# inputs need: country_sector_df, world_sector_df, country_all_df, world_all_df

# have: mx_xports_sector_10, world_xports_sector_10, mx_xports_all, world_xports_all

rca_sector_12 <- calculate_rca_generalized(mx_xports_sector_12, world_xports_sector_12,mx_xpo
```

Plot sector 12

Visualize over time:

```r
## ggplot to bisualize over time
ggplot(rca_sector_12, aes(x = period, y = RCA, color = as.factor(cmdCode), group = cmdCode))
  geom_line(size = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed") +
  labs(title = "RCA Over Time - HS12 Sector Products (12)",
       x = "Year", y = "RCA") +
  theme_bw()
```



13 Shellac; gum, fat and other plant liquids, juice

```
mx_xports_sector_13 <- read_csv("mx_xports_prod_i/mx_xports_sector_13.csv")

world_xports_sector_13 <- read_csv("world_xports_prod_i/world_xports_sector_13.csv")

mx_xports_sector_13 <- necessary_variables(mx_xports_sector_13)
world_xports_sector_13 <- necessary_variables(world_xports_sector_13)
```

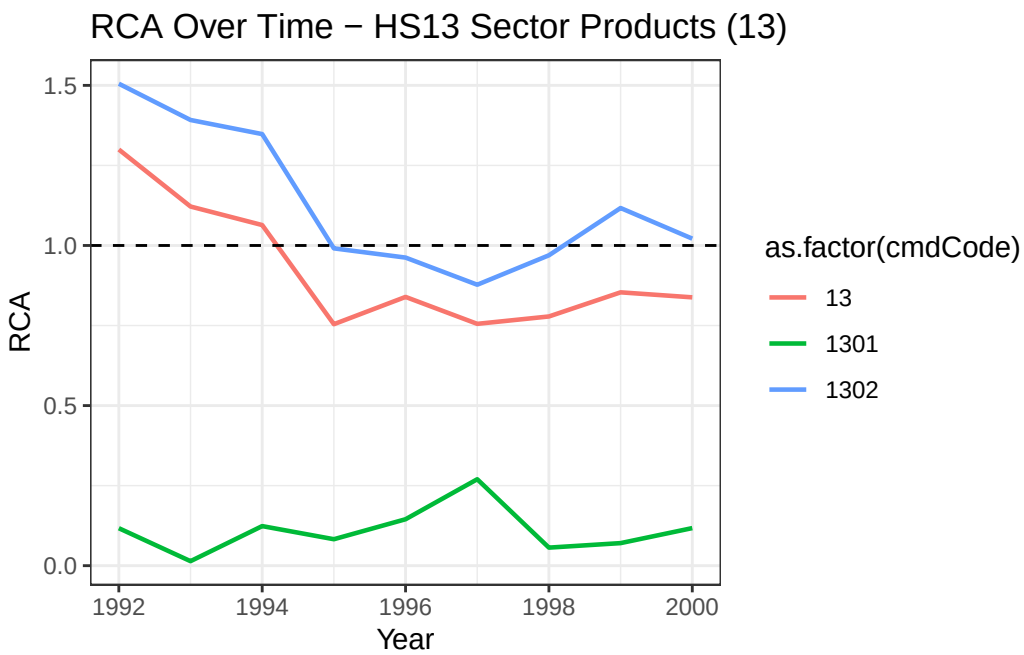
RCA Sector 13

```
# inputs need: country_sector_df, world_sector_df, country_all_df, world_all_df

# have: mx_xports_sector_10, world_xports_sector_10, mx_xports_all, world_xports_all

rca_sector_13 <- calculate_rca_generalized(mx_xports_sector_13, world_xports_sector_13,mx_xpo
```

Plot sector 13

Visualize over time:

```r
## ggplot to bisualize over time
ggplot(rca_sector_13, aes(x = period, y = RCA, color = as.factor(cmdCode), group = cmdCode))
  geom_line(size = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed") +
  labs(title = "RCA Over Time - HS13 Sector Products (13)",
       x = "Year", y = "RCA") +
  theme_bw()
```



14 Planting materials for planting; other plant products

```r
mx_xports_sector_14 <- read_csv("mx_xports_prod_i/mx_xports_sector_14.csv")

world_xports_sector_14 <- read_csv("world_xports_prod_i/world_xports_sector_14.csv")

mx_xports_sector_14 <- necessary_variables(mx_xports_sector_14)
world_xports_sector_14 <- necessary_variables(world_xports_sector_14)
```

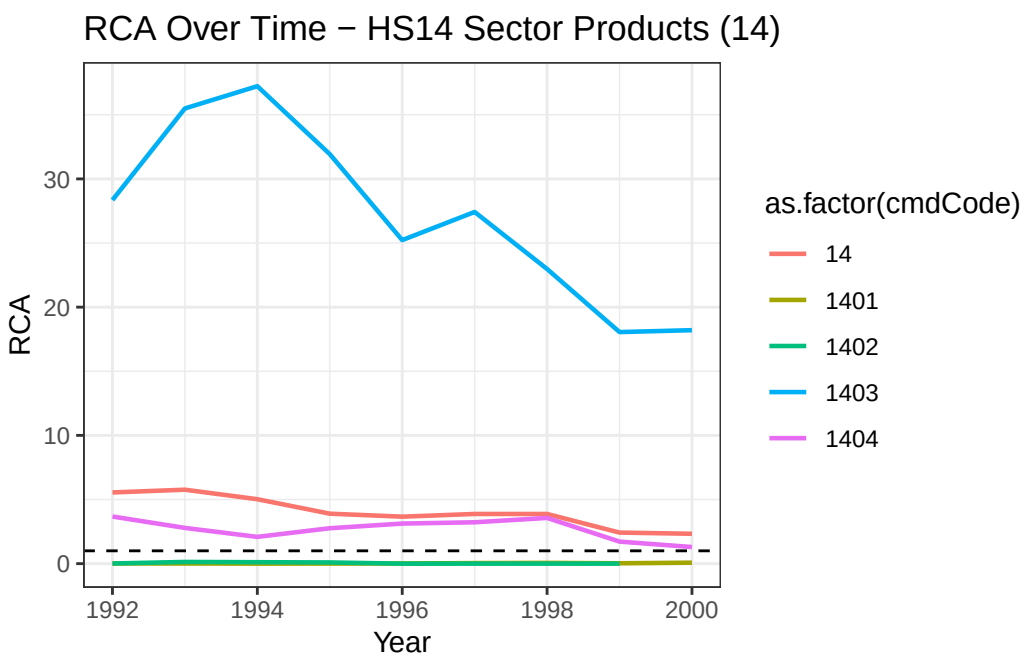
RCA Sector 14

```
# inputs need: country_sector_df, world_sector_df, country_all_df, world_all_df

# have: mx_xports_sector_10, world_xports_sector_10, mx_xports_all, world_xports_all

rca_sector_14 <- calculate_rca_generalized(mx_xports_sector_14, world_xports_sector_14,mx_xpo
```

Plot sector 14

Visualize over time:

```
## ggplot to bisualize over time
ggplot(rca_sector_14, aes(x = period, y = RCA, color = as.factor(cmdCode), group = cmdCode))
  geom_line(size = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed") +
  labs(title = "RCA Over Time - HS14 Sector Products (14)",
       x = "Year", y = "RCA") +
  theme_bw()
```



15 Animal and vegetable oils and fats; refined edible oils and fats

We Omit the following:

- 1503
- 1505
- 1506
- -1508
- 1510
- 1511 to 1514
- 1517 to 1519

```
mx_xports_sector_15 <- read_csv("mx_xports_prod_i/mx_xports_sector_15.csv")

world_xports_sector_15 <- read_csv("world_xports_prod_i/world_xports_sector_15.csv")

mx_xports_sector_15 <- necessary_variables(mx_xports_sector_15)
world_xports_sector_15 <- necessary_variables(world_xports_sector_15)
```

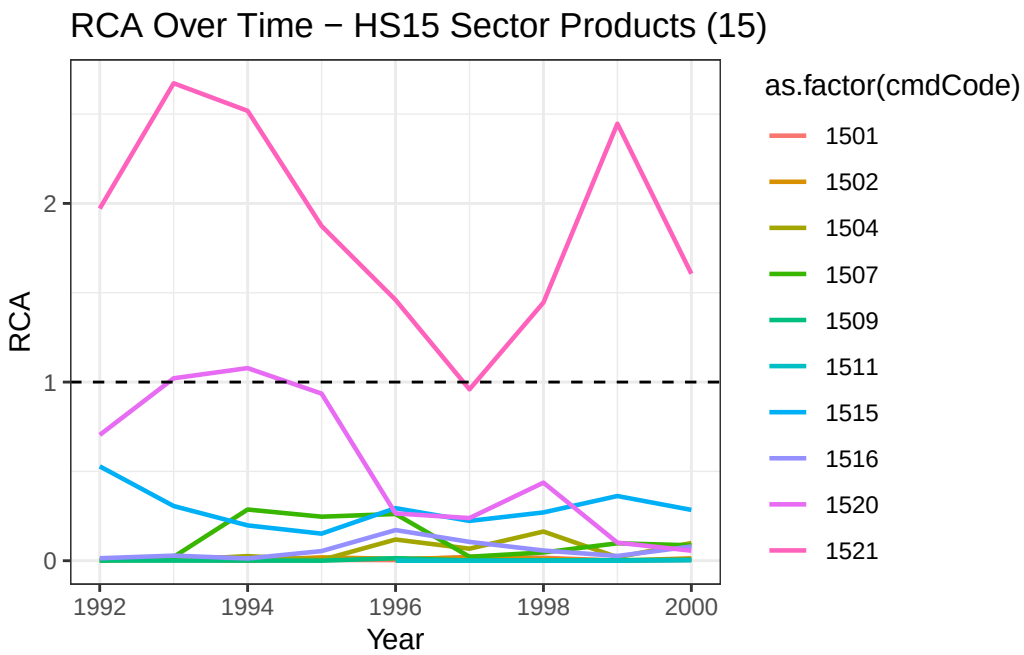
RCA Sector 15

```
# inputs need: country_sector_df, world_sector_df, country_all_df, world_all_df

# have: mx_xports_sector_10, world_xports_sector_10, mx_xports_all, world_xports_all

rca_sector_15 <- calculate_rca_generalized(mx_xports_sector_15, world_xports_sector_15,mx_xpo
```

Plot sector 15

Visualize over time:

```
## ggplot to bisualize over time
ggplot(rca_sector_15, aes(x = period, y = RCA, color = as.factor(cmdCode), group = cmdCode))
  geom_line(size = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed") +
  labs(title = "RCA Over Time - HS15 Sector Products (15)",
       x = "Year", y = "RCA") +
  theme_bw()
```



16 Meat, fish and other aquatic invertebrate products

RCA Sector 16

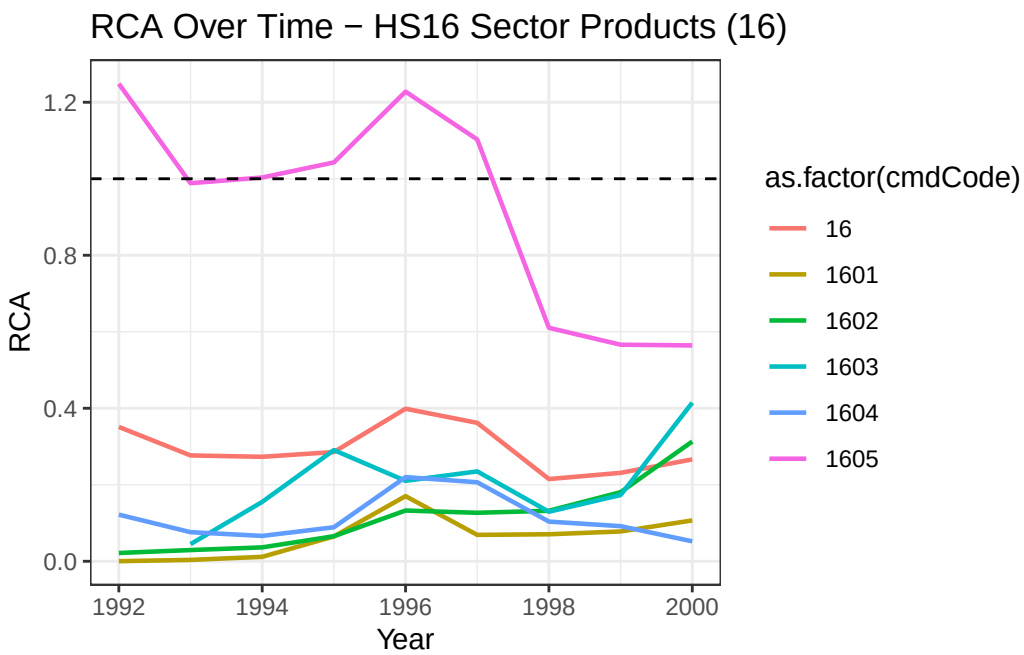
```
mx_xports_sector_16 <- read_csv("mx_xports_prod_i/mx_xports_sector_16.csv")

world_xports_sector_16 <- read_csv("world_xports_prod_i/world_xports_sector_16.csv")

mx_xports_sector_16 <- necessary_variables(mx_xports_sector_16)
world_xports_sector_16 <- necessary_variables(world_xports_sector_16)
```

```
# inputs need: country_sector_df, world_sector_df, country_all_df, world_all_df

# have: mx_xports_sector_10, world_xports_sector_10, mx_xports_all, world_xports_all

rca_sector_16 <- calculate_rca_generalized(mx_xports_sector_16, world_xports_sector_16,mx_xpo
```

Plot sector 16

Visualize over time:

```r
## ggplot to bisualize over time
ggplot(rca_sector_16, aes(x = period, y = RCA, color = as.factor(cmdCode), group = cmdCode))
  geom_line(size = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed") +
  labs(title = "RCA Over Time - HS16 Sector Products (16)",
       x = "Year", y = "RCA") +
  theme_bw()
```



17 Sugar and confectionery

RCA Sector 17

```r
mx_xports_sector_17 <- read_csv("mx_xports_prod_i/mx_xports_sector_17.csv")

world_xports_sector_17<- read_csv("world_xports_prod_i/world_xports_sector_17.csv")

mx_xports_sector_17 <- necessary_variables(mx_xports_sector_17)
world_xports_sector_17 <- necessary_variables(world_xports_sector_17)
```

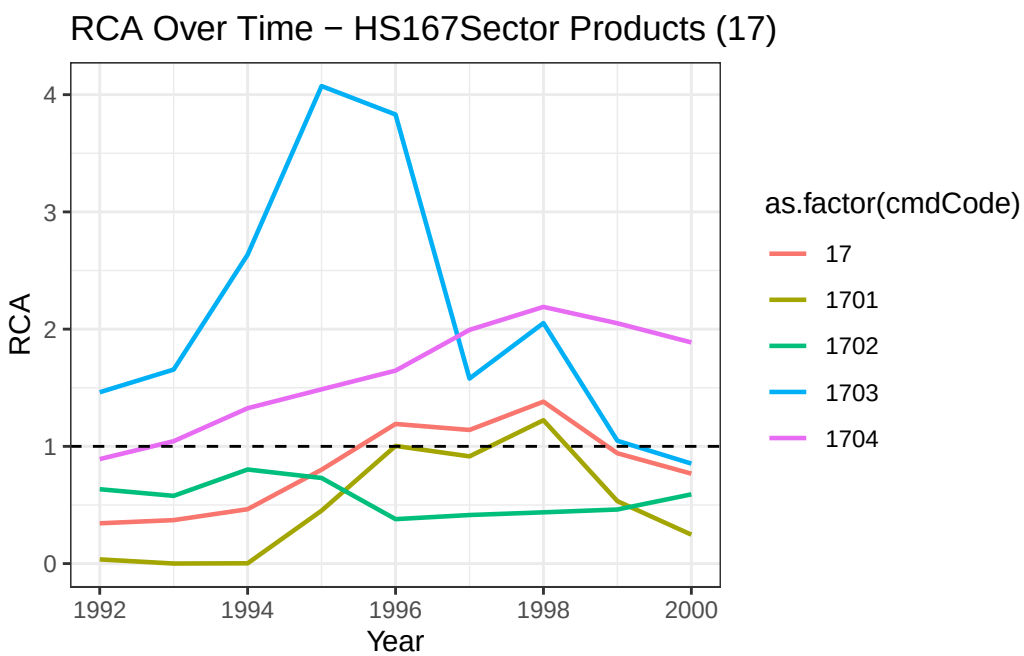
```r
# inputs need: country_sector_df, world_sector_df, country_all_df, world_all_df
```

```
# have: mx_xports_sector_10, world_xports_sector_10, mx_xports_all, world_xports_all

rca_sector_17 <- calculate_rca_generalized(mx_xports_sector_17, world_xports_sector_17,mx_xpo
```

Plot sector 16

Visualize over time:

```
## ggplot to bisualize over time
ggplot(rca_sector_17, aes(x = period, y = RCA, color = as.factor(cmdCode), group = cmdCode))
  geom_line(size = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed") +
  labs(title = "RCA Over Time - HS167Sector Products (17)",
       x = "Year", y = "RCA") +
  theme_bw()
```



18 Cocoa and cocoa products

RCA Sector 18

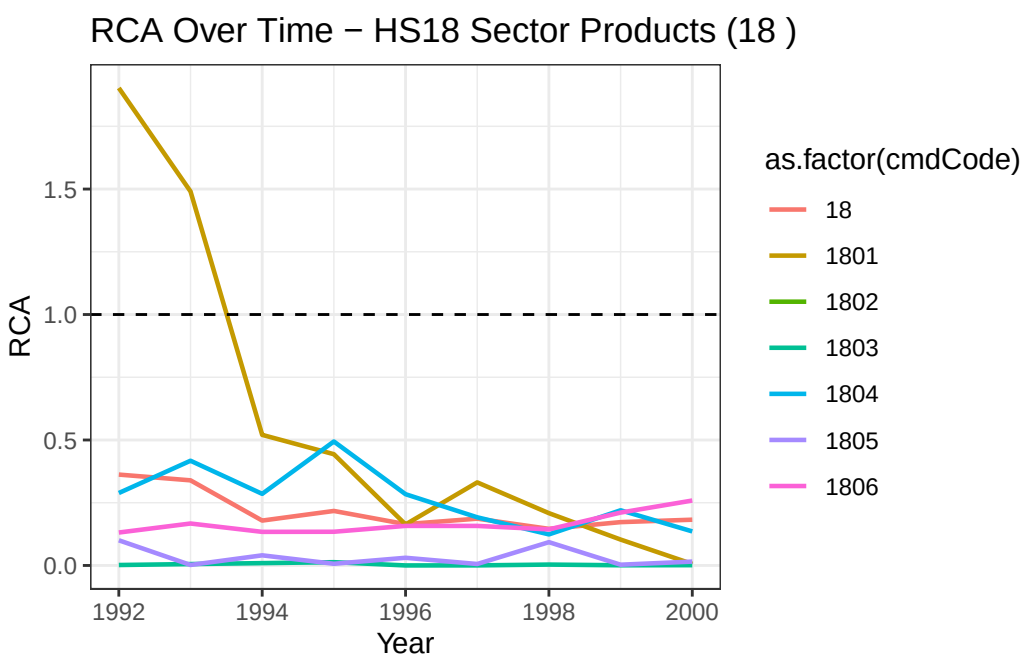
```r
mx_xports_sector_18 <- read_csv("mx_xports_prod_i/mx_xports_sector_18.csv")

world_xports_sector_18 <- read_csv("world_xports_prod_i/world_xports_sector_18.csv")

mx_xports_sector_18 <- necessary_variables(mx_xports_sector_18)
world_xports_sector_18 <- necessary_variables(world_xports_sector_18)
```

```r
# inputs need: country_sector_df, world_sector_df, country_all_df, world_all_df

# have: mx_xports_sector_10, world_xports_sector_10, mx_xports_all, world_xports_all

rca_sector_18 <- calculate_rca_generalized(mx_xports_sector_18, world_xports_sector_18,mx_xpo
```

Plot sector 18

Visualize over time:

```r
## ggplot to bisualize over time
ggplot(rca_sector_18, aes(x = period, y = RCA, color = as.factor(cmdCode), group = cmdCode))
  geom_line(size = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed") +
  labs(title = "RCA Over Time - HS18 Sector Products (18 )",
       x = "Year", y = "RCA") +
  theme_bw()
```



19 Cereal flour, starch, etc. or dairy products; cakes

20 Products of vegetables, fruits or other parts of plants

21 Miscellaneous food

22 Beverages, wine and vinegar

23 Food industry residues and waste; formulated feed

24 Tobacco, tobacco and tobacco substitute products

50 Silk

51 Wool and other animal hair

52 Cotton

53 Other plant fibers

Miscellaneous and Extra Code

```
##GENERALIZED RCA ANALYSIS

## Generalized RCA function for any product code
#calculate_rca_for_product <- function(product_code) {

  #  build file paths for  product
 # mx_prod_path <- paste0("mx_xports_prod_i/mx_xports_", product_code, ".csv")
 # world_prod_path <- paste0("world_xports_prod_i/world_xports_", product_code, ".csv")

  # load and read the  data
  #mx_xports_prod <- read_csv(mx_prod_path, show_col_types = FALSE)
 # world_xports_prod <- read_csv(world_prod_path, show_col_types = FALSE)
```

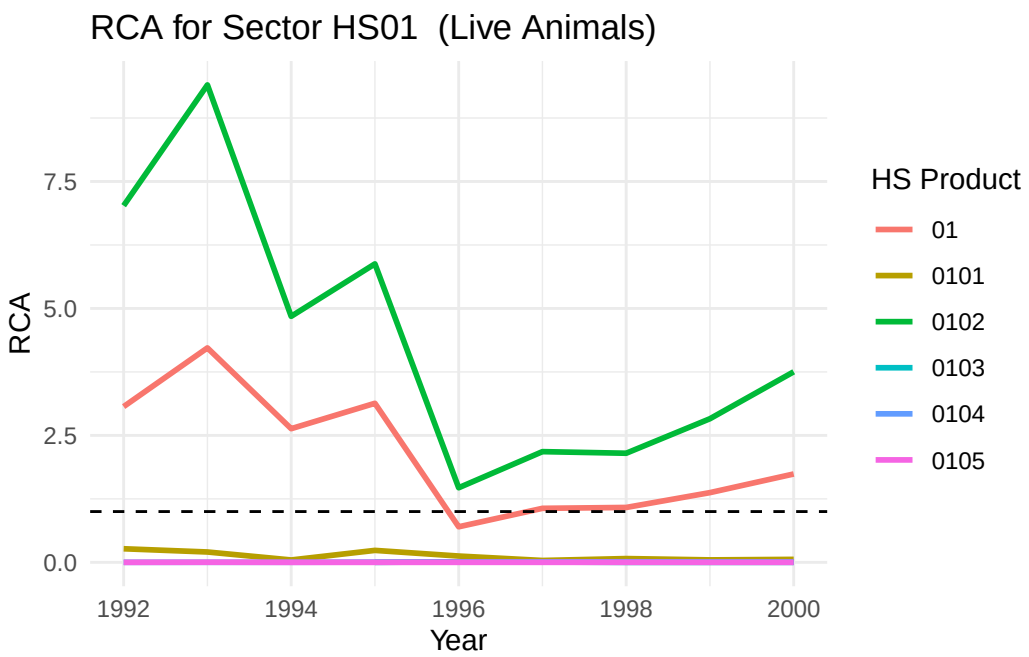
```
  # laod world totals for time frame
  #mx_xports_all <- read_csv("mx_total_xports_world/mx_xports_all.csv", show_col_types = FALS
  #world_xports_all <- read_csv("world_total_xports/world_xports_all.csv", show_col_types = F

  # clean and arrange using  necessary_varioables function
#  mx_xports_prod <- necessary_variables(mx_xports_prod)
#  world_xports_prod <- necessary_variables(world_xports_prod)
#  mx_xports_all <- necessary_variables(mx_xports_all)
#  world_xports_all <- necessary_variables(world_xports_all)

  # calculate RCA
#  rca_df <- mx_xports_prod %>%
 #   inner_join(mx_xports_all, by = "period", suffix = c("_prod", "_total")) %>%
#    inner_join(world_xports_prod, by = "period") %>%
 #   inner_join(world_xports_all, by = "period", suffix = c("_world_prod", "_world_total")) %
 #   mutate(
   #   RCA = (fobvalue_prod / fobvalue_total) /
  #          (fobvalue_world_prod / fobvalue_world_total)
 #   ) %>%
  #  select(period, RCA) %>%
 #   mutate(product_code = product_code)

 # return(rca_df)
#}
```

run RCA all sectors

```
rca_all <- purrr::map_dfr(sector_codes, process_sector)

## keep only first two digits

rca_all <- rca_all |>
  mutate(
    cmdCode = as.character(cmdCode),
    sector = substr(cmdCode, 1, 2)
  )
```

plot one sector (look at HS01 Animals: live for refernece)

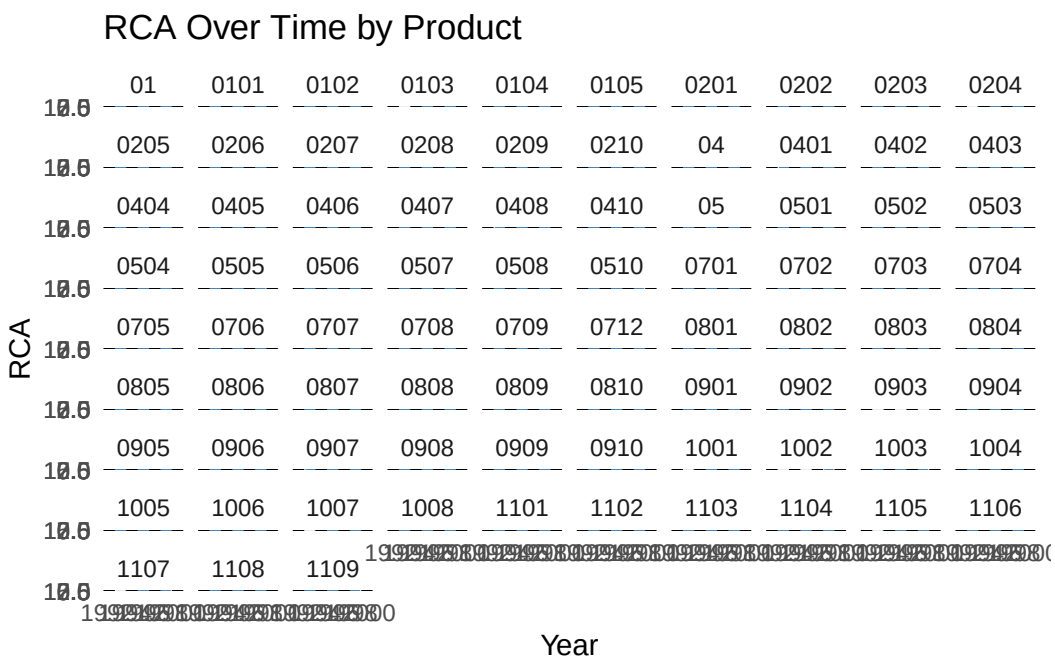
```
ggplot(
  rca_all |> filter(sector == "01"), #filter for first sector
  aes(x = period, y = RCA, color = cmdCode, group = cmdCode)
) +
  geom_line(size = 1) +
  geom_hline(yintercept = 1, linetype = "dashed") +
  labs(
    title = "RCA for Sector HS01  (Live Animals)",
    x = "Year",
    y = "RCA",
    color = "HS Product"
  ) +
  theme_minimal()
```



plot all sectors at once

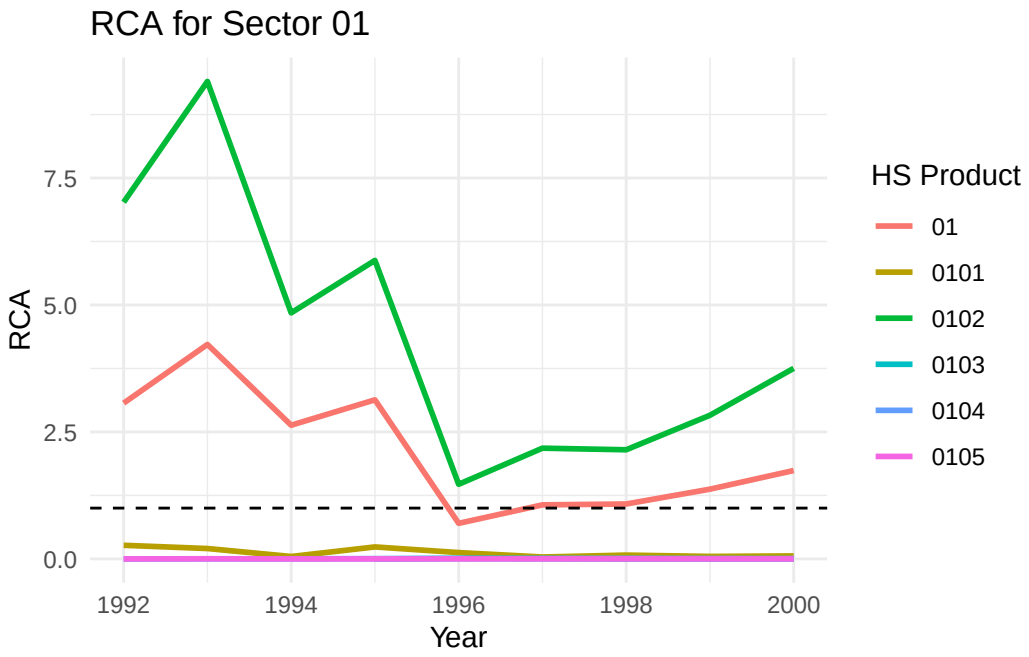
```
ggplot(rca_all, aes(x = period, y = RCA)) +
  geom_line(color = "steelblue") +
```

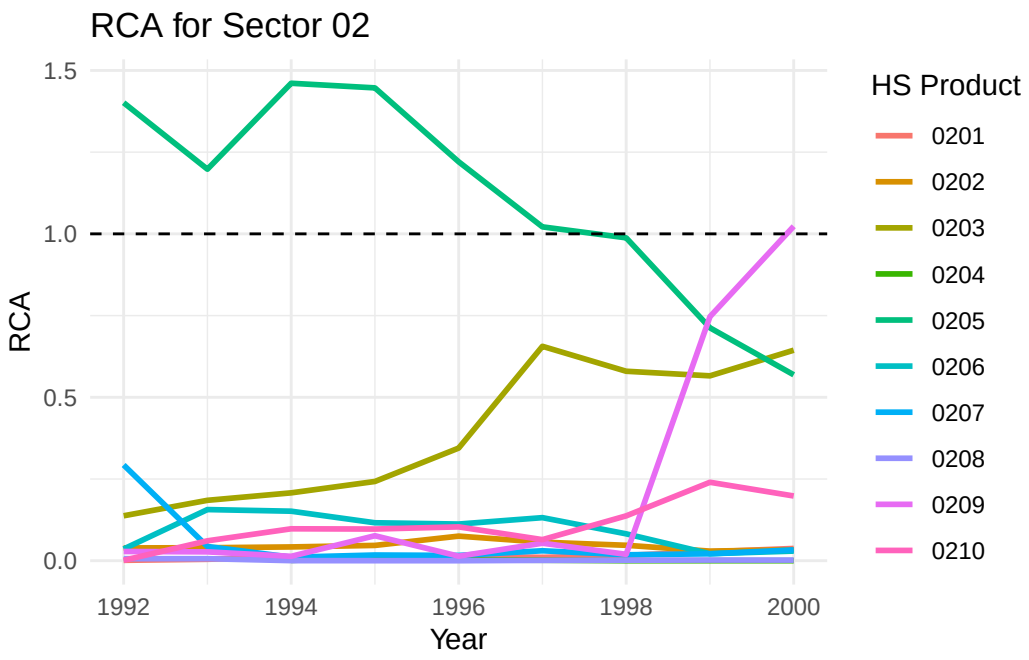
```
  geom_hline(yintercept = 1, linetype = "dashed") +
  facet_wrap(~cmdCode) +
  theme_minimal() +
  labs(
    title = "RCA Over Time by Product",
    x = "Year",
    y = "RCA"
  )
```

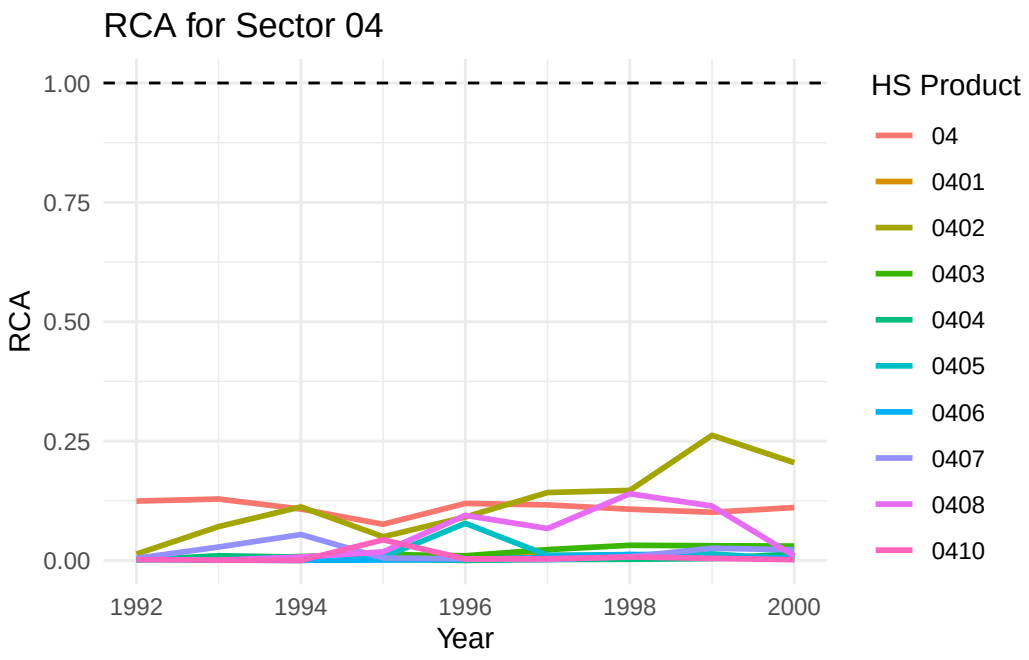


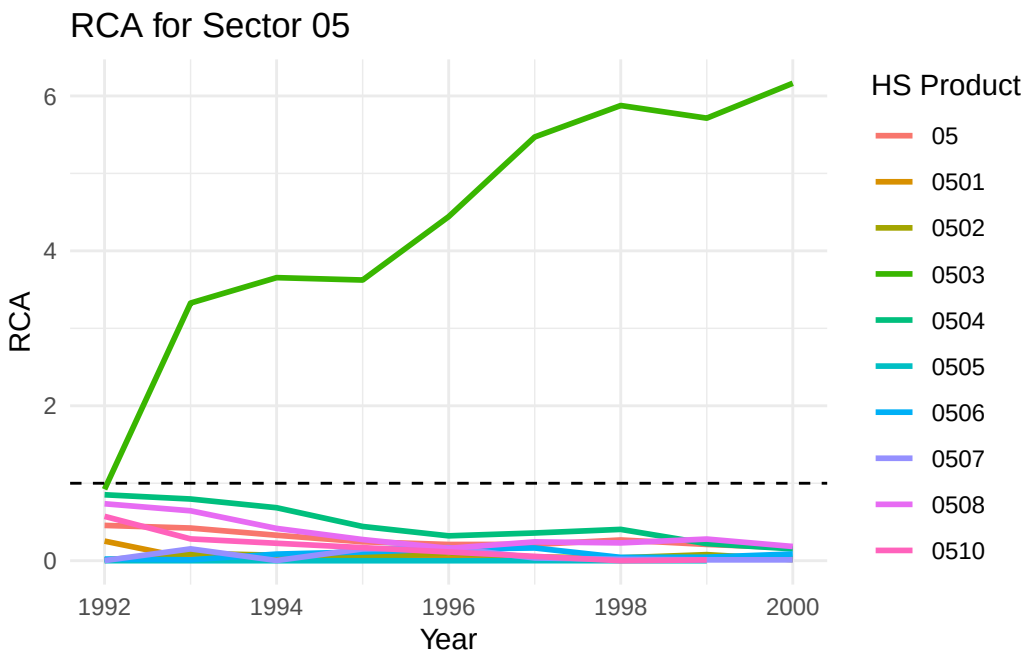

```
sectors <- unique(rca_all$sector) ## creatrs a list for all elements in sector_code MAKE SURE


for (s in sectors) { ## for all elements s in list sectors,

  allplots <- ggplot(
    rca_all |> filter(sector == s), ## filter for HS code loops
    aes(x = period, y = RCA, color = cmdCode, group = cmdCode)
  ) +
    geom_line(size = 1) +
    geom_hline(yintercept = 1, linetype = "dashed") +
    labs(
      title = paste("RCA for Sector", s),
      x = "Year",
```

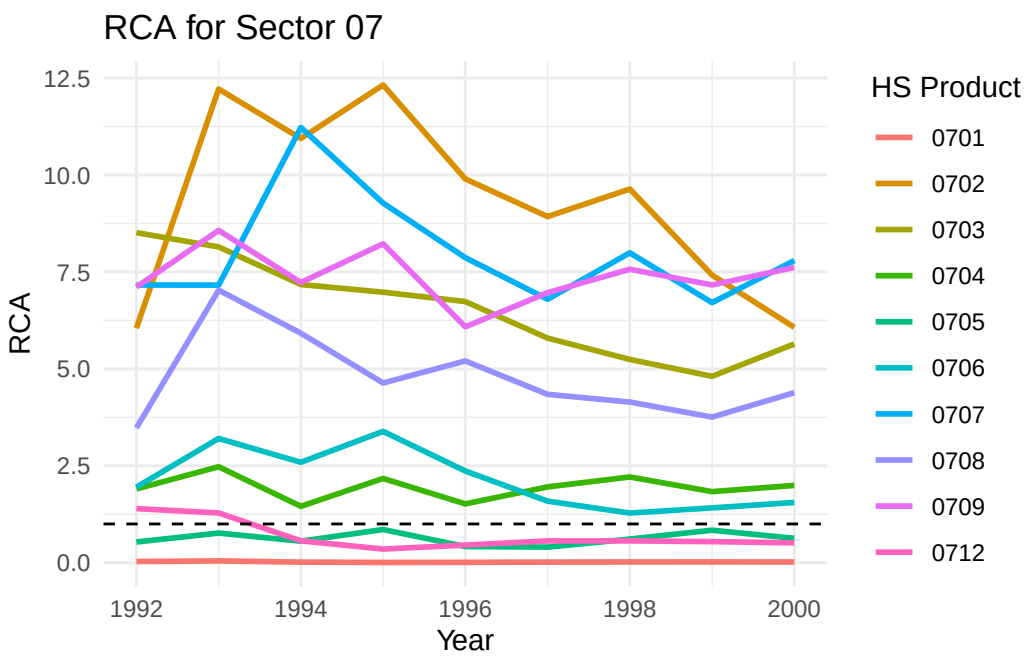
```
      y = "RCA",
      color = "HS Product"
    ) +
    theme_minimal()

  print(allplots)
}
```

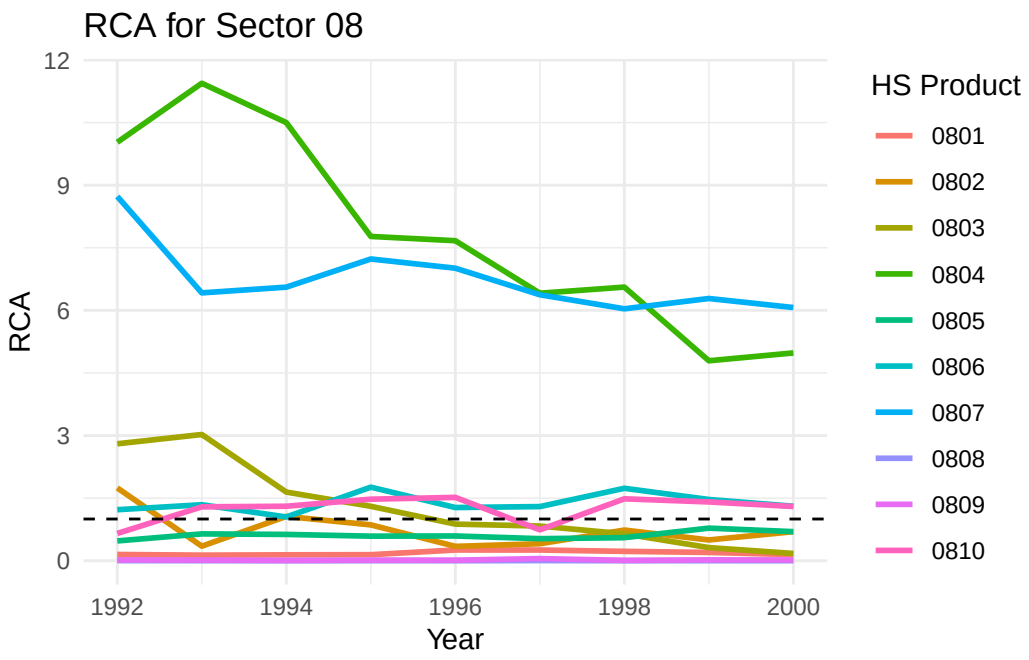


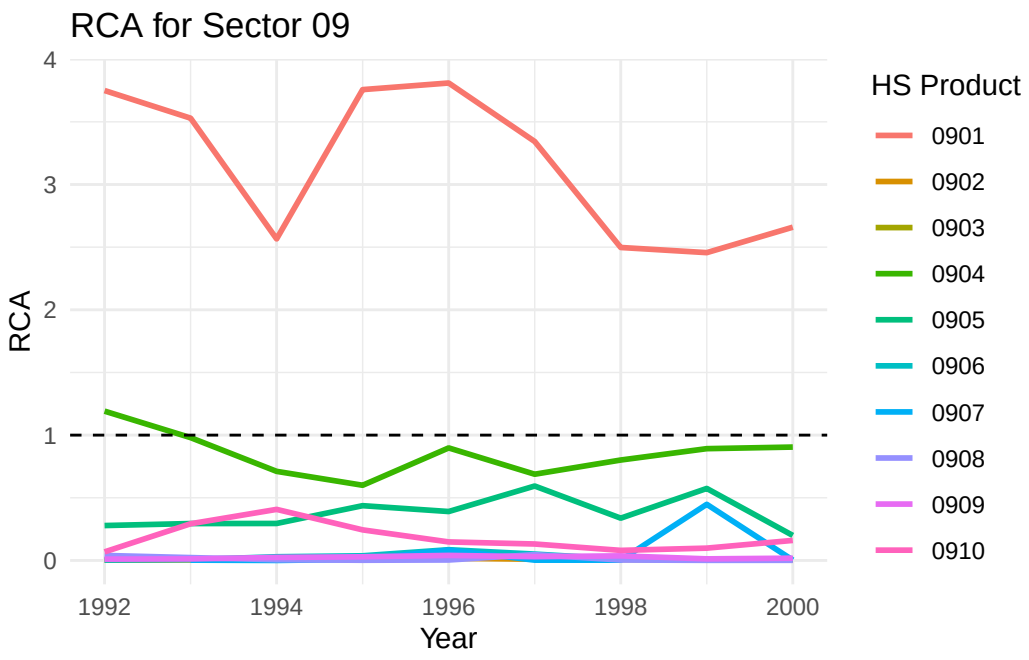

RCA for Sector 02
1.5
1.0
0.5
0.0
RCA
1992
1994
1996
1998
2000
Year
HS Product
0201
0202
0203
0204
0205
0206
0207
0208
0209
0210

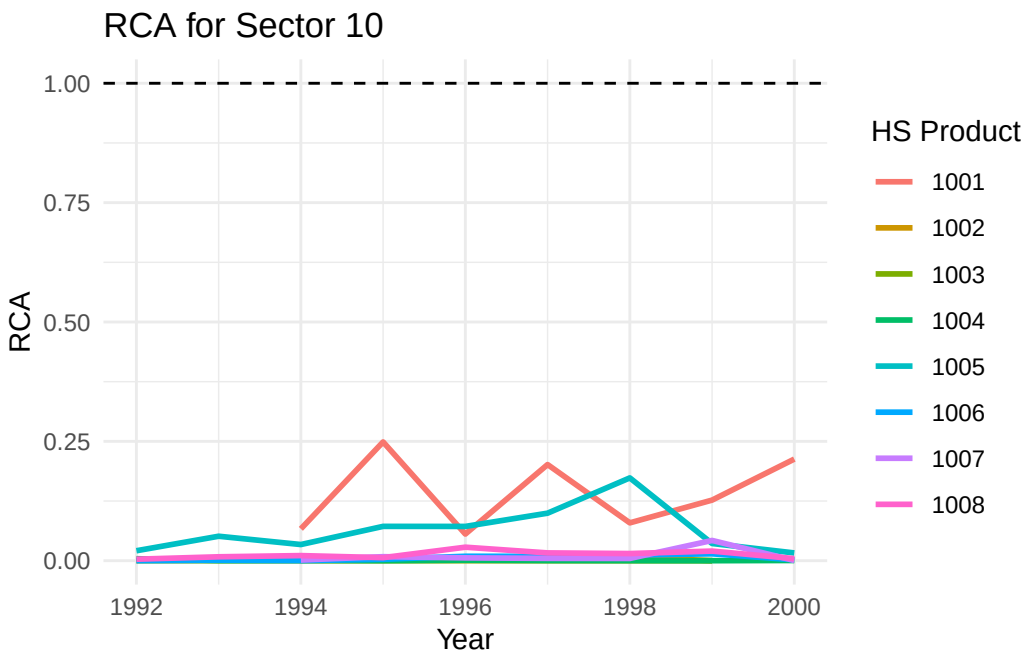

RCA for Sector 04
1.00
0.75
0.50
0.25
0.00
RCA
1992
1994
1996
1998
2000
Year
HS Product
04
0401
0402
0403
0404
0405
0406
0407
0408
0410

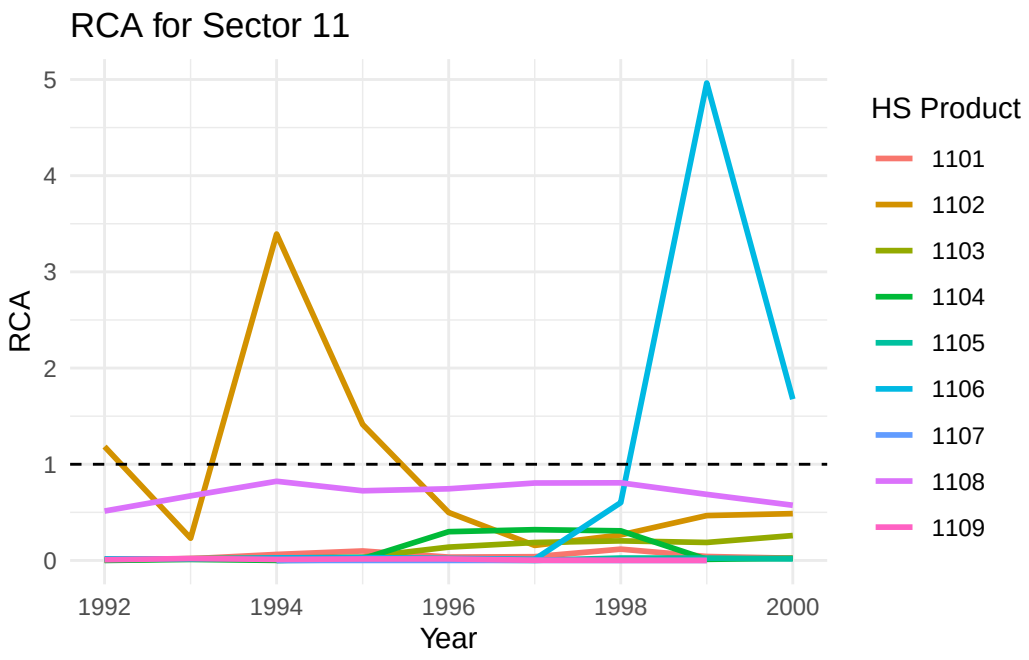

RCA for Sector 05
RCA
Year
HS Product
05
0501
0502
0503
0504
0505
0506
0507
0508
0510


RCA for Sector 07
RCA
Year
HS Product
0701
0702
0703
0704
0705
0706
0707
0708
0709
0712


RCA for Sector 08
12
9
6
3
0
RCA
1992
1994
1996
1998
2000
Year
HS Product
0801
0802
0803
0804
0805
0806
0807
0808
0809
0810


RCA for Sector 09
4
3
2
1
0
RCA
1992
1994
1996
1998
2000
Year
HS Product
0901
0902
0903
0904
0905
0906
0907
0908
0909
0910


RCA for Sector 10
1.00
0.75
0.50
0.25
0.00
RCA
1992
1994
1996
1998
2000
Year
HS Product
1001
1002
1003
1004
1005
1006
1007
1008


RCA for Sector 11
5
4
3
2
1
0
RCA
1992
1994
1996
1998
2000
Year
HS Product
1101
1102
1103
1104
1105
1106
1107
1108
1109